DECK 07 · WEEK 4 · AFTERNOON 1

FSA II — Cash Flow, Inventories & LT Assets

Three accounting choices that quietly drive earnings quality, tax, and comparability

CFA Level I · Vol 4 Learning Modules 4–7

Roadmap — FSA II: Cash Flow, Inventories & Long-Lived Assets

5 sections · ~2 hours · Built bottom-up from CFA readings

01

Cash flow statement · structure & preparation

3-bucket CFS · direct vs indirect · indirect→direct conversion · investing & financing build-up

02

Common-size CF + FCFF / FCFE

% inflows vs % revenue · FCFF from CFO and from NI · FCFE · 6 performance + 6 coverage ratios

03

Inventory methods & ratio impact

LCNRV · FIFO / LIFO / weighted-avg · inflation effects · LIFO reserve · LIFO liquidation · disclosures

04

PP&E · capitalize vs expense · depreciation

capitalize vs expense · SL / DDB / units · useful-life sensitivity · component depreciation · revaluation

05

Intangibles · impairment · R&D

purchased / internal / M&A · goodwill · impairment IFRS vs GAAP · derecognition · R&D IFRS vs GAAP

Learning objectives

By the end of this deck you should be able to:

1. Prepare operating cash flow under both the direct and indirect methods, reconcile them, and convert from one to the other using balance sheet and income statement data.

2. Compute FCFF and FCFE from both CFO and net income, and interpret cash-flow performance and coverage ratios.

3. Explain how FIFO, LIFO, and weighted-average cost flow assumptions — and the lower-of-cost-and-NRV rule — affect the income statement, balance sheet, and ratios under IFRS and US GAAP.

4. Capitalize or expense an expenditure on a long-lived asset, choose an appropriate depreciation method, and quantify the sensitivity of earnings to depreciation estimates.

5. Account for intangibles by origin, test for impairment under IFRS vs US GAAP, and apply the distinct IFRS and US GAAP rules on research and development costs.

SECTION 01

Cash flow statement structure & preparation

The statement of cash flows reconciles opening and closing cash by sorting every movement into operating, investing, or financing

Ending cash = beginning cash + CFO + CFI + CFF + FX effects. Every financing decision ultimately shows up in cash.

$$\Delta \text{Cash} = \text{CFO} + \text{CFI} + \text{CFF} \quad (+ \text{ FX effects on cash balances})$$

OPERATING — CFO

Cash from day-to-day business: receipts from customers, payments to suppliers, wages, taxes. The main engine of long-run value creation.

INVESTING — CFI

Cash to acquire or dispose of long-lived assets: PP&E, intangibles, and financial instruments held for investment. Capex lives here.

FINANCING — CFF

Cash to/from capital providers: debt issuance/repayment, equity issuance/buyback, dividends paid. IFRS allows more latitude than US GAAP.

Source: CFA Institute, Curriculum Vol 4, Learning Module 4, §1–2, pp. 125–131.

Direct and indirect methods produce the same CFO total — they only differ in how operating cash flow is presented

IFRS encourages the direct method; US GAAP permits either, but 99%+ of filers use indirect. Analysts often convert back.

DIRECT METHOD — ACME 2018

Cash received from customers	23,543
Cash paid to suppliers	(11,900)
Cash paid to employees	(4,113)
Cash paid for other opex	(3,532)
Cash paid for interest	(258)
Cash paid for income tax	(1,134)
Net cash from operating activities	2,606

INDIRECT METHOD — ACME 2018

Net income	2,210
+ Depreciation	1,052
– Gain on sale of equipment	(205)
– Δ Accounts receivable	(55)
– Δ Inventory	(707)
+ Δ Accounts payable	263
+ other working-capital adjustments	48
Net cash from operating activities	2,606

Source: CFA Institute, Curriculum Vol 4, Learning Module 4, §3–4, pp. 131–141.

Any indirect-method CFO can be re-expressed in direct format via a three-step process — a core analyst skill

The approximation is accurate enough for most analytical purposes. Key insight: every accrual item maps to a specific cash line.

Cash received from customers = Revenue − Δ Accounts receivable

CFA LM 4 — ACME CORP. 2018 CONVERSION

Starting data: Revenue 23,598; COGS 11,456; salaries 4,123; other opex 3,577; interest 246; tax 1,139. ΔAR +55, ΔInv +707, ΔAP +263, ΔPrepaid −23, ΔSalaryPay +10, ΔIntPay −12, ΔTaxPay +5, ΔAccrued +22.

1. Step 1 — Aggregate: total revenue 23,803; total expenses 21,593; NI = 2,210.
2. Step 2 — Remove non-cash/non-operating: back out the 205 gain on equipment sale (revenue side) and 1,052 depreciation (expense side).
3. Step 3 — Adjust accruals for ΔWC to get cash: customers 23,598−55 = 23,543; suppliers 11,456+707−263 = 11,900; employees 4,123−10 = 4,113; other opex 3,577−23−22 = 3,532.
4. Finish: interest 246+12 = 258; tax 1,139−5 = 1,134. Sum to USD 2,606.

Direct CFO = 23,543 − 11,900 − 4,113 − 3,532 − 258 − 1,134 = USD 2,606 ✓

Source: CFA Institute, Curriculum Vol 4, Learning Module 4, §5, pp. 141–142.

A full CFS walks from net income to ending cash through operating, investing and financing, and ties to the balance sheet

Investing and financing are always presented on a gross, "direct" basis — regardless of which method is used for CFO.

Line	Acme 2018	Section
Net cash from operating activities	2,606	CFO
Cash received from sale of equipment	762	CFI
Cash paid for purchase of equipment	(1,300)	CFI
Net cash used in investing activities	(538)	CFI
Cash paid to retire long-term debt	(500)	CFF
Cash paid to retire common stock	(600)	CFF
Cash paid for dividends	(1,120)	CFF
Net cash used in financing activities	(2,220)	CFF
Net decrease in cash (2,606 – 538 – 2,220)	**(152)**	**Check**

BALANCE-SHEET TIE-OUT

Opening cash 1,163 – 152 = 1,011 ending cash — matches 12/31/2018 balance sheet exactly.

Source: CFA Institute, Curriculum Vol 4, Learning Module 4, §2–4, pp. 131–141.

Investing and financing sections are always presented directly — and purchases and sales of PP&E require careful disaggregation

Net equipment change on the balance sheet conflates purchases with disposals. The income-statement gain gives you the third equation.

Cash from sale = Book value of asset sold + Gain on sale

where: Book value = historical cost − accumulated depreciation, each computed from the BS roll-forward.

CFA LM 4 — COPPER INC. EQUIPMENT DISPOSAL

BS: Equipment 100 → 109; Accum. dep. 30 → 36. Capex USD 15m (all new equipment). IS: depreciation 8m, gain on sale 12m. Cash from sale?

1. Historical cost of equipment sold: 100 + 15 − 109 = USD 6m
2. Accum. depreciation on equipment sold: 30 + 8 − 36 = USD 2m
3. Book value of equipment sold: 6 − 2 = USD 4m
4. Cash received from sale: book value + gain = 4 + 12 = USD 16m

Cash inflow in CFI = USD 16m. CFI net = +16 − 15 = +USD 1m from equipment transactions.

Source: CFA Institute, Curriculum Vol 4, Learning Module 4, §6 Question Set, pp. 143–144.

SECTION 02

Common-size & free cash flow

Common-size cash flow analysis scales each line either to total inflows/outflows or to net revenue — both reveal different trends

Inflow/outflow view highlights mix of sources and uses; revenue view aids forecasting (each item as a % of sales).

Acme 2018 cash flow line	Amount (USD)	% inflows / outflows	% net revenue
Cash received from customers	23,543	96.86%	99.77%
Cash paid to suppliers	(11,900)	48.66%	50.43%
Cash paid to employees	(4,113)	16.82%	17.43%
Cash paid for other opex	(3,532)	14.44%	14.97%
Net cash from operating activities	2,606	—	11.04%
Capital expenditure	(1,300)	5.32%	5.51%
Dividends paid	(1,120)	4.58%	4.75%

WHY BOTH MATTER

Inflow/outflow view: is CFO the dominant source of cash? Revenue view: does each item scale with sales at the expected ratio?

Source: CFA Institute, Curriculum Vol 4, Learning Module 5, §2–3, pp. 157–164.

FCFF and FCFE separate the cash available to all capital providers from the cash available to equity holders alone

FCFF discounts at WACC to value the firm. FCFE discounts at cost of equity to value equity directly. Consistent discount-rate matching is non-negotiable.

$$FCFF = NI + NCC + Int \times (1 - t) - FCInv - WCInv$$

$$FCFE = CFO - FCInv + Net\ borrowing$$

where: NCC = non-cash charges (mostly D&A). FCInv = capex. WCInv = Δ non-cash working capital. Net borrowing = new debt issued − debt repaid.

FCFF — FIRM LEVEL

Cash available to ALL capital providers (debt + equity) after reinvestment in fixed and working capital. Add back interest × (1−t) because debt-holders are paid BEFORE this line.

FCFE — EQUITY LEVEL

Cash available to common stockholders after interest, principal repayment and reinvestment. Do NOT add back after-tax interest; DO adjust for net borrowing.

Source: CFA Institute, Curriculum Vol 4, Learning Module 5, §4, pp. 167–168.

A worked example — Acme's FCFE from net income — shows why the path matters more than the destination

Tax rate here is 34% (= tax expense / pretax income = 1,139 / 3,349). Two paths — NI-based and CFO-based — must reconcile exactly.

CFA LM 5 — ACME CORP. FCFF & FCFE FROM CFO

CFO = 2,606. Capex = 1,300. Proceeds from equipment sale = 762. Interest paid = 258. Net debt repayment = 500. Tax rate t = 34%.

1. FCInv (net) = 1,300 − 762 = USD 538
2. FCFF (from CFO) = CFO + Int×(1−t) − FCInv = 2,606 + 258×0.66 − 538 = 2,606 + 170 − 538 = USD 2,238
3. FCFE = CFO − FCInv − Net debt repayment = 2,606 − 538 − 500 = USD 1,568
4. Cross-check (FCFE from NI): NI 2,210 + NCC 1,052 − FCInv 538 − WCInv ≈ (1,568) and ± net borrowing (0 here). Reconciles ✓

FCFF = USD 2,238 · FCFE = USD 1,568 — a healthy 60% of FCFF flows to equity.

INTERPRETATION

Positive FCFE means Acme can sustainably pay dividends or buy back stock. Acme's actual dividend payment (1,120) sits comfortably below FCFE (1,568) — a coverage ratio of 1.40×.

Source: CFA Institute, Curriculum Vol 4, Learning Module 5, §4 Exhibit 8, p. 168.

Cash flow ratios split cleanly into performance ratios (profitability) and coverage ratios (solvency) — all computed from CFO

Cash-based ratios sidestep accrual manipulation. They are especially useful for comparing earnings quality across issuers.

Ratio	Formula	What it measures	Type
Cash flow to revenue	CFO / Net revenue	Cash margin per dollar of sales	Perf.
Cash return on assets	CFO / Avg. total assets	Cash per dollar of assets	Perf.
Cash return on equity	CFO / Avg. shareholders' equity	Cash per dollar of equity	Perf.
Cash to income	CFO / Operating income	Earnings quality	Perf.
Debt coverage	CFO / Total debt	Financial risk / leverage	Cover.
Interest coverage	(CFO + Int + Tax) / Interest paid	Can the company service debt?	Cover.
Reinvestment	CFO / Capex	Can CFO fund capex?	Cover.
Dividend payment	CFO / Dividends paid	Can CFO fund dividends?	Cover.

Source: CFA Institute, Curriculum Vol 4, Learning Module 5, §5 Exhibit 9, pp. 168–170.

Performance ratios in practice — Microsoft's CFO/revenue and CFO/assets reveal earnings quality Apple has lost since 2015

Both are strong businesses. But trend matters: Microsoft's CFO/revenue is rising, Apple's is falling — a subtle earnings-quality signal.

Company	CFO / Revenue 2015	2016	2017	CFO / Assets 2015	2016	2017
Microsoft	31.7%	39.1%	43.9%	17.1%	18.1%	18.2%
Apple Inc.	34.8%	30.5%	27.7%	31.1%	21.5%	18.2%

WHAT THE NUMBERS SAY

1. CFO/revenue — Microsoft now generates 43.9¢ of operating cash per dollar of sales, vs 27.7¢ at Apple. Microsoft's +12 pp trend over 3 years vs Apple's −7 pp is a classic earnings-quality divergence.

2. CFO/assets — Apple's drop from 31% → 18% reflects enormous treasury buildup: CFO divided by a growing investment portfolio. The underlying operating business remains cash-rich; the ratio mechanics hide it.

3. Analyst takeaway — never read a ratio in isolation. Decompose, question, and benchmark against the firm's own 5-year history before drawing conclusions.

Source: CFA Institute, Curriculum Vol 4, Learning Module 5, Example 3, p. 170.

SECTION 03

Inventory methods & ratio impact

Under IFRS inventory is carried at the lower of cost and NRV; US GAAP uses lower of cost / market and — for LIFO — forbids reversals

NRV = estimated selling price − costs to complete − costs to sell. Write-downs hit COGS; reversals (IFRS only) reduce COGS in the recovery year.

IFRS carrying amount = MIN (historical cost, NRV)

Scenario	IFRS	US GAAP (LIFO filer)
Cost 5.2m; NRV declines to 4.7m	Write down to 4.7m; 0.5m in COGS	Write down to 4.7m; 0.5m in COGS
Prior-year NRV recovers from 4.7m back to 5.0m	Reverse up to 5.0m; 0.3m reduces COGS	No reversal allowed (stuck at 4.7m)
Cost 5.2m; replacement cost 4.9m, NRV 5.1m	Use NRV ⇒ no write-down	Use lower of cost / market ⇒ 4.9m
Inventory pledged as security	Disclose carrying amount	Disclose carrying amount

WHY IT MATTERS

Write-downs reduce current-period net income but improve future gross margin when the inventory is sold. Reversals under IFRS partially offset this.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, §2, pp. 175–181.

FIFO, LIFO, and weighted-average allocate the same cost pool differently between COGS and ending inventory — identical cash, different profit

The physical flow of goods is irrelevant to the accounting choice. LIFO is not permitted under IFRS; weighted-average and FIFO are.

FIFO

First-in, First-out

Oldest costs → COGS
Most-recent costs → ending inventory

BS closest to replacement cost
IS mismatches old COGS vs current sales

In inflation: low COGS, high profit, high tax

LIFO (US GAAP only)

Last-in, First-out

Most-recent costs → COGS
Oldest costs → ending inventory

IS closest to replacement cost
BS inventory at stale, low costs

In inflation: high COGS, low profit, LOW TAX

WEIGHTED AVERAGE

Single blended cost

Avg cost = total cost / total units

Every unit (COGS + BS) at the same cost

Between FIFO and LIFO on every metric

Widely used in continental Europe and Asia

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, §3, pp. 181–184.

In a rising-price environment LIFO raises COGS and lowers ending inventory — cutting reported income and taxes but distorting ratios

Companies L and F are identical except for the cost-flow assumption. 4% annual price inflation; 10% unit-volume growth; 2,000-unit base.

Year	Ending inv. (L-LIFO)	Ending inv. (F-FIFO)	COGS (L)	COGS (F)	Gross profit (L)	Gross profit (F)
1	16,000	16,000	160,000	160,000	140,000	140,000
2	16,000	16,640	183,040	182,400	160,160	160,800
3	16,000	17,306	209,398	208,732	183,223	183,889
4	16,000	17,998	239,551	238,859	209,607	210,299
5	16,000	18,718	274,046	273,326	239,791	240,511
5-yr	**0**	**+2,718 (F > L)**	**+3,472 (L >**	**—**	**-3,472 (L < F)**	**—**

TAX IMPACT IF THEY WERE US FILERS AT 25% TAX RATE

Over 5 years, Company L reports USD 3,472 less gross profit than Company F — a pure tax shield from LIFO.

Cash tax saving ≈ 3,472 × 25% = USD 868 kept inside the firm. This is LIFO's entire economic rationale.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, Example 3, pp. 182–184.

The LIFO reserve bridges a US-LIFO filer back to a comparable FIFO view — indispensable for cross-border comparison

LIFO filers disclose the LIFO reserve in the footnotes. Adding it back gives the FIFO-equivalent carrying amount — directly comparable to IFRS filers.

$$\text{Inv_FIFO} = \text{Inv_LIFO} + \text{LIFO reserve} \qquad \text{COGS_FIFO} = \text{COGS_LIFO} - \Delta\ \text{LIFO reserve}$$

where: Δ LIFO reserve = ending − beginning. Positive Δ → inventory cost rising → LIFO COGS > FIFO COGS by exactly that amount.

Line (JPY millions)	As reported (LIFO)	LIFO reserve	Adjusted (FIFO)
Ending inventory	608,572	+10,120	618,692
Beginning inventory (prior year)	594,200	+6,380	600,580
Avg inventory (for turnover)	601,386	+8,250	609,636
COGS	3,160,000	− Δ reserve 3,740	3,156,260
Inventory turnover (COGS/avg inv.)	**5.25×**		**5.18×**

BOTTOM LINE

FIFO-adjusted turnover is always LOWER than LIFO-reported turnover in rising-price environments — do not mistake this for worse inventory management.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, §4 Question Set, pp. 186–195.

A LIFO liquidation — selling more than is replenished — releases old low-cost layers into COGS, inflating gross profit unsustainably

A declining inventory balance in a LIFO filer is a red flag. Check the 10-K for "LIFO liquidation" disclosure — required under US GAAP.

STYLISED EXAMPLE — LIFO LIQUIDATION

A firm holds 2,000 base-year units at USD 8/unit cost (total USD 16,000). Current cost is USD 14/unit. In 2024 the firm sells 2,500 units but only purchases 2,000.

1. Normal COGS: 2,500 × 14 = USD 35,000 (if all units came from 2024 purchases at current cost).
2. Actual LIFO COGS: 2,000 × 14 (current year layer) + 500 × 8 (base-year layer) = 28,000 + 4,000 = USD 32,000.
3. Liquidation gain: 35,000 − 32,000 = USD 3,000 = 500 units × (14 − 8) spread between current and base-year cost.
4. Gross profit boost: USD 3,000 of one-time, unsustainable gross margin uplift — with a matching inventory balance drop of USD 4,000.

Analyst adjustment: strip out the USD 3,000 from gross profit when assessing sustainable margin.

MANAGEMENT INCENTIVES

LIFO liquidations can also be triggered deliberately: running down inventory to boost reported EPS in a weak quarter. The liquidation REVERSES the tax benefit the LIFO reserve was holding — so the earnings boost is also a cash-tax hit.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, §3, p. 183.

Inventory disclosures let analysts recompute turnover, days-on-hand and gross margin on a consistent basis across companies

The required IFRS disclosures — policy, carrying amount by class, write-down, reversals — are the raw material for inventory-quality analysis.

Ratio	Before adjustment	After removing write-down	Interpretation
Gross profit margin	23.95%	23.89%	Slightly lower w/o w-down
Net profit margin	6.36%	6.31%	Tax offsets most of diff.
Inventory turnover (× p.a.)	4.18×	4.16×	Slightly more efficient w/ w-down

ANALYST CHECKLIST

1. Extract inventory method (policy note) — if LIFO, get LIFO reserve to restate on FIFO.
2. Reconcile write-downs and reversals to identify trends.
3. Compare inventory growth vs revenue growth — if inventory outpaces sales, risk of obsolescence.
4. Benchmark inventory turnover and DOH against 3 industry peers, not just the firm's own history.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, Example 4 (Jollof Inc.), pp. 186–195.

A worked lower-of-cost-and-NRV example highlights the asymmetric treatment of reversals under IFRS vs US GAAP

Two years: a write-down of EUR 0.5m in 2023; a partial recovery to EUR 5.0m in 2024. Same economics, different earnings outcome.

HATSUMEI-STYLE WORKED EXAMPLE

2023: original cost EUR 5.2m; NRV declines to EUR 4.7m. 2024: NRV recovers to EUR 5.0m (still below original cost of 5.2m).

1. Under IFRS 2023: write-down 5.2 → 4.7; recognise EUR 0.5m loss in COGS. Carrying amount 4.7.
2. Under IFRS 2024: reversal 4.7 → 5.0; recognise EUR 0.3m reduction in COGS. Carrying amount 5.0. Note: cap at original cost.
3. Under US GAAP 2023 (LIFO filer): write-down 5.2 → 4.7; 0.5m in COGS. Carrying amount 4.7.
4. Under US GAAP 2024: NO REVERSAL. Carrying amount stays at 4.7 even though NRV is 5.0.

IFRS 2-yr net impact: -0.2m. US GAAP 2-yr net impact: -0.5m. US GAAP is structurally more conservative.

COMPARABILITY ADJUSTMENT

When benchmarking a US GAAP LIFO filer against an IFRS peer, add back any permanent-but-recovered write-downs captured in the US firm's financials — otherwise its gross margin is artificially depressed in the recovery years.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, §2, pp. 175–181.

SECTION 04

PP&E · capitalize vs expense · depreciation

Capitalizing spreads a cost across future periods; expensing hits earnings today — the choice reshapes income, assets, and leverage ratios

Capitalize if the expenditure provides future economic benefit > 1 year and can be reliably measured. Otherwise, expense as incurred.

Dimension	Capitalize	Expense
Year-1 net income	Higher (only dep'n hits)	Lower (full amount hits)
Years 2+ net income	Lower (ongoing depreciation)	Higher (no further charge)
Total assets	Higher (asset on BS)	Lower (no asset recorded)
Shareholders' equity	Higher (smoothed earnings in retained)	Lower in year 1, higher later
CFO	Higher (outflow in CFI)	Lower (outflow in CFO)
CFI	Lower (capex outflow)	Unchanged
Debt-to-equity	Lower (equity higher)	Higher in year 1
ROA year 1	Capex ÷ bigger base → lower	No impact on asset side

AGGRESSIVE ACCOUNTING RED FLAG

If a company capitalizes costs that peers expense, margins look artificially high and ROA artificially low — look for rapid growth in "other non-current assets" or "capitalized software".

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §2, pp. 211–216.

Straight-line is the default depreciation method; DDB front-loads expense; units-of-production matches expense to actual use

All three methods depreciate the SAME total amount (cost − salvage) over the life — they differ ONLY in the timing of the charge.

STRAIGHT-LINE

Dep_t = (Cost − Salvage) / Useful life

Flat annual charge

Default under IFRS and US GAAP; most common.

Use when benefits are reasonably even over the life (office buildings, most furniture).

DOUBLE-DECLINING BALANCE

Dep_t = (2 / Useful life) × NBV_(t−1)

Accelerated — highest expense in year 1

Do NOT subtract salvage in the formula; simply stop once NBV reaches salvage.

Use when productivity drops with age (vehicles, certain machinery, tech equipment).

UNITS OF PRODUCTION

Dep_t = (Cost − Salvage)
× (Units used / Total units)

Activity-based — matches wear to output

Use for assets where consumption is measurable: mine output, aircraft cycles, printing-press impressions.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §2, pp. 213–216.

Worked example — USD 50,000 asset, 5-year life, USD 5,000 salvage — shows how method choice shifts the depreciation curve

DDB rate = 2 / 5 = 40%. Applied to opening NBV. Switch to SL whenever it produces a higher charge (standard convention).

Year	SL expense	SL net book value	DDB expense	DDB net book value
1	9,000	41,000	20,000	30,000
2	9,000	32,000	12,000	18,000
3	9,000	23,000	7,200	10,800
4	9,000	14,000	4,320	6,480
5	9,000	5,000	1,480	5,000
Total	**45,000**	**Matches salvage**	**45,000**	**Matches salvage**

NET INCOME IMPACT AT 25% TAX RATE

Year 1: SL reduces pretax income by 9,000 → tax 2,250; DDB reduces it by 20,000 → tax 5,000. DDB saves USD 2,750 of cash tax in year 1.

Over 5 years total depreciation is identical (45,000) — so is total tax. DDB is a timing play: earlier deductions at the cost of higher later earnings.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §2 — SL vs DDB illustration, pp. 213–216.

Useful life, salvage value, and depreciation method are three estimates — small changes produce large effects on reported earnings

Extending useful life from 5 to 8 years cuts annual depreciation by 38% — a popular "earnings management" lever auditors watch carefully.

Assumption change	Annual SL depreciation	Δ vs baseline	Year-5 NBV
Baseline: life 5y, salvage 5,000	9,000	—	5,000
Life extended to 8y (same salvage)	5,625	−3,375 (−38%)	21,875
Salvage raised to 10,000 (life still 5y)	8,000	−1,000 (−11%)	10,000
Switch to DDB (life 5y, salvage 5,000)	20,000 yr 1	+11,000 yr 1	5,000
Both: life 8y AND salvage 10,000	5,000	−4,000 (−44%)	25,000

HOW TO VET ESTIMATES

Compare the firm's stated useful lives against peers (required disclosure under IFRS). Watch for unexplained upward revisions after a weak quarter. A simultaneous increase in useful life AND salvage is a double-barrelled earnings boost.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §2, pp. 213–216.

IFRS requires component depreciation for material parts with different useful lives; US GAAP permits but rarely applies it

Common example: an aircraft with a 30-year airframe, 10-year engine, and 8-year interior — three components, three depreciation schedules.

COMPONENT DEPRECIATION — IFRS

Each material component is depreciated separately using its own useful life and residual value.

Consequences:
· More detailed fixed-asset register
· Earlier, larger write-offs on shorter-life parts
· Avoids overstating asset life on a blended basis

Required under IAS 16 when component cost is significant relative to the total.

REVALUATION MODEL — IFRS ONLY

Alternative to the cost model. Revalue PP&E to fair value periodically.

· Revaluation surplus → OCI (not P&L)
· Depreciation then runs on the new, higher base
· A subsequent decline reverses the OCI first, then hits P&L

US GAAP: revaluation PROHIBITED for PP&E. Historical cost only.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §4, pp. 223–225.

SECTION 05

Intangibles · impairment · R&D

Intangible assets are accounted for differently by origin: purchased, internally developed, or acquired in a business combination

Goodwill exists only from a business combination — it is the residual after all identifiable assets and liabilities are valued at fair value.

PURCHASED SEPARATELY

e.g. a single patent or domain name bought from a third party.

· Recorded at fair value = purchase price
· Finite-life: amortise; test for impairment when triggers
· Indefinite-life (rare): no amortisation; annual impairment test

INTERNALLY DEVELOPED

Generally EXPENSE as incurred.

Exceptions:
· IFRS: capitalize DEVELOPMENT costs once criteria met
· US GAAP: capitalize certain software development costs

Creates a structural asymmetry vs purchased intangibles.

ACQUIRED IN M&A

Allocate the purchase price to each asset at fair value.

· Identifiable intangibles get their own line (patents, brands, customer lists)
· Any residual is GOODWILL
· Goodwill is not amortised — tested annually for impairment

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §2, pp. 213–217.

Impairment writes an asset down when carrying amount exceeds recoverable amount — IFRS and US GAAP use different tests

Under IFRS the write-down can be smaller (value-in-use often > fair value) but reversible. Under US GAAP it can be larger and is usually permanent.

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IFRS: loss  =  CA – MAX (FV – costs to sell, Value-in-use)
US GAAP: if CA > undisc. CF, loss  =  CA – FV
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where: CA = carrying amount. Value-in-use = PV of future cash flows. US GAAP is a two-step test; IFRS is a single step.

CFA LM 7 EXAMPLE 3 — ESSEX MACHINE

Carrying amount GBP 18,000. Undiscounted CFs GBP 16,000. PV of CFs (value-in-use) GBP 14,000. Fair value if sold GBP 10,000. Costs to sell GBP 2,000.

1. IFRS recoverable amount = MAX(10,000 − 2,000; 14,000) = 14,000
2. IFRS impairment loss = 18,000 − 14,000 = GBP 4,000
3. US GAAP Step 1: compare CA 18,000 vs undisc. CF 16,000 → CA is NOT recoverable → step 2
4. US GAAP Step 2: loss = CA − FV = 18,000 − 10,000 = GBP 8,000

IFRS loss GBP 4,000 · US GAAP loss GBP 8,000 — same machine, USD 4,000 earnings difference.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §3 Examples 2–3, pp. 217–221.

Derecognition removes an asset from the balance sheet — gain or loss equals sale proceeds minus carrying amount

Under the indirect method, the gain is REMOVED from CFO (to avoid double counting) and the full proceeds appear in CFI.

Gain / (loss) on sale = Sale proceeds − (Historical cost − Accum. depreciation)

where: Carrying amount at time of sale = historical cost − accumulated depreciation − any prior impairment.

CFA LM 7 EXAMPLE 4 — MOUSSILAUKE DINERS SELLS 450 PIZZA OVENS

Historical cost USD 5.1m. Accumulated depreciation USD 3.2m. Sale proceeds USD 3.1m.

1. Carrying amount: 5.1 − 3.2 = USD 1.9m
2. Gain on sale: 3.1 − 1.9 = USD 1.2m
3. Income statement: USD 1.2m gain (typically in "Other gains")
4. CFS (indirect): subtract 1.2m from CFO; show 3.1m as CFI inflow.

Gain = USD 1.2m. Net CFI impact = +USD 3.1m (all proceeds sit in investing activities).

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §3 Example 4, pp. 220–221.

IFRS permits capitalization of development costs once criteria are met; US GAAP expenses R&D except certain software

IFRS criteria: technical feasibility + intent to complete + ability to use/sell + future benefits + reliable measurement + adequate resources.

CFA LM 7 EXAMPLE 1 — REH AG SOFTWARE DEVELOPMENT

REH AG (IFRS filer) spends EUR 1,000 per month over FY2019 to develop software for internal use. Annual total EUR 12,000.

1. Criteria met on 1 Feb 2019: Jan costs EUR 1,000 EXPENSED. Feb–Dec EUR 11,000 CAPITALIZED as intangible asset.
2. Criteria met on 1 Dec 2019: Jan–Nov EUR 11,000 EXPENSED. Dec EUR 1,000 CAPITALIZED.
3. Alternative under US GAAP with probability established in 2018: full EUR 12,000 CAPITALIZED (internal-use software).
4. Both regimes: prior-period expenses cannot be retroactively reclassified to an asset — once expensed, always expensed.

IFRS date-sensitive, US GAAP probability-sensitive. Analyst must monitor the "capitalized development costs" line year-over-year for consistency.

ANALYTICAL IMPACT

A company that capitalizes more R&D looks more profitable AND more asset-heavy than a peer that expenses it. For consistent comparison, analysts commonly expense ALL R&D (the "conservative adjustment") across peers.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §2 Example 1, pp. 213–215.

Goodwill is the residual in an acquisition — not amortised, but tested annually for impairment; large goodwill is a common analyst red flag

AB InBev / SABMiller 2016: USD 103.1bn purchase price → USD 74.1bn of goodwill. Impairment test = compare CGU's recoverable amount to its carrying amount including goodwill.

AB InBev purchase-price allocation	USD million	% of price
Property, plant & equipment	9,060	8.8%
Intangible assets (brands, customer lists, etc.)	20,040	19.4%
Inventories, receivables, cash & other assets	9,117	8.8%
Assets held for sale (to be divested)	24,805	24.1%
Less: total liabilities + non-controlling interests	(33,969)	(32.9)%
Net identifiable assets acquired	29,053	28.2%
Goodwill (purchase price – net identifiable assets)	74,083	71.8%
Total purchase consideration	**103,136**	**100.0%**

WHY GOODWILL MATTERS

72% of the price is goodwill — any future impairment hits earnings directly. Goodwill-heavy deals warrant close monitoring of the acquired unit's cash flows.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §2 Exhibit 1 (AB InBev / SABMiller), pp. 216–217.

KEY TAKEAWAYS

Deck 7 — FSA II: Cash Flow, Inventories & Long-Lived Assets

- CFO is identical under direct and indirect methods; indirect → direct is a three-step process (aggregate → remove non-cash → adjust ΔWC). Acme 2018 CFO = USD 2,606 either way.
- Cash-flow ratios: 4 performance ratios (CFO/revenue, /assets, /equity, /operating income) + 4 coverage ratios (debt, interest, reinvestment, dividend coverage).
- LCNRV: write-down hits COGS. Reversals allowed under IFRS, forbidden under US GAAP — a source of persistent earnings-comparability issues.
- Impairment: IFRS recoverable = MAX(FV−sell, VIU); US GAAP = 2-step undisc-CF then FV. Essex machine: IFRS loss GBP 4,000 vs US GAAP loss GBP 8,000.
- FCFF = NI + NCC + Int(1−t) − FCInv − WCInv = CFO + Int(1−t) − FCInv. FCFE = CFO − FCInv + Net borrowing. Match FCFF with WACC, FCFE with cost of equity.
- In rising prices, LIFO → higher COGS, lower income, lower tax, lower ending inventory; FIFO → the reverse. LIFO reserve bridges LIFO to FIFO: Inv_FIFO = Inv_LIFO + LIFO reserve.
- Straight-line is flat (9,000 p.a. on the 50k/5y/5k example). DDB front-loads (20,000 in year 1). Both depreciate the same TOTAL but time differently.
- R&D: IFRS capitalises development costs once criteria met; US GAAP expenses all R&D except certain software. Goodwill arises only in M&A, is not amortised, and is tested annually.